

Real Analysis HW 8

1. (Ch 6. 33) Let $\{f_n\}$ be a sequence of real-valued functions on $[a, b]$ that converges pointwise on $[a, b]$ to the real-valued function f . Show that

$$TV(f) \leq \liminf_{n \rightarrow \infty} TV(f_n).$$

Pf. 任取 $\varepsilon > 0$. 由 $\liminf$ 的性质, 存在子列 $\{f_{k_n}\}$ 使得

$$TV(f_{k_n}) < \liminf_{n \rightarrow \infty} TV(f_n) + \frac{\varepsilon}{2}.$$

对一个固定的分割 $P: x_0 = a < x_1 < \dots < x_m = b$, 由于 $\{f_{k_n}\}$ 点点收敛到 f , 且 f 点点有限, 存在 $N(\varepsilon, k)$, 当 $n > N(\varepsilon, k)$ 时,

$$|f_{k_n}(x_k) - f(x_k)| < \frac{\varepsilon}{4m}$$

全

$$N(\varepsilon) = \max_{1 \leq k \leq m} \{N(\varepsilon, k)\}$$

则 $n > N(\varepsilon)$ 时, 对一切 $1 \leq k \leq m$ 有

$$|f(x_k) - f(x_{k-1})| < |f_{k_n}(x_k) - f_{k_n}(x_{k-1})| + 2 \cdot \frac{\varepsilon}{4m}.$$

于是

$$\sum_{k=1}^m |f(x_k) - f(x_{k-1})| < \sum_{k=1}^m |f_{k_n}(x_k) - f_{k_n}(x_{k-1})| + \frac{\varepsilon}{2}$$

即得

$$V(f, P) < V(f_{k_n}, P) + \frac{\varepsilon}{2} \leq TV(f_{k_n}) + \frac{\varepsilon}{2} < \liminf_{n \rightarrow \infty} TV(f_n) + \varepsilon.$$

由于 P 已固定, 与 ε 无关, 令 $\varepsilon \rightarrow 0^+$ 得

$$V(f, P) \leq \liminf_{n \rightarrow \infty} TV(f_n).$$

最后对 P 取 $\sup$ 得

$$TV(f) = \sup_{P \text{ a partition}} \{V(f, P)\} \leq \liminf_{n \rightarrow \infty} TV(f_n)$$

□.
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2. (Ch 6. 34). Let f and g be of bounded variation on $[a, b]$. Show that

$$TV(f+g) \leq TV(f) + TV(g)$$

and

$$TV(\alpha f) = |\alpha| TV(f).$$

Pf. 对一个固定的分割 $P: x_0 = a < x_1 < \dots < x_m = b$. 由三角不等式

$$\begin{aligned} V(f+g, P) &= \sum_{k=1}^m |(f+g)(x_k) - (f+g)(x_{k-1})| \\ &= \sum_{k=1}^m |(f(x_k) - f(x_{k-1})) + (g(x_k) - g(x_{k-1}))| \\ &\leq \sum_{k=1}^m |f(x_k) - f(x_{k-1})| + \sum_{k=1}^m |g(x_k) - g(x_{k-1})| \\ &= V(f, P) + V(g, P) \\ &\leq TV(f) + TV(g). \end{aligned}$$

对 P 取 $\sup$ 即证第一个不等式. 另一方面

$$\begin{aligned} TV(\alpha f) &= \sup_{P \text{ a partition}} \left| \sum_{k=1}^m \alpha f(x_k) - \alpha f(x_{k-1}) \right| \\ &= |\alpha| \sup_{P \text{ a partition}} \left| \sum_{k=1}^m (f(x_k) - f(x_{k-1})) \right| \\ &= |\alpha| TV(f). \end{aligned}$$

□.

3. (Ch 6. 37). Let f be a continuous function on $[0, 1]$ that is absolutely continuous on $[\varepsilon, 1]$ for each $0 < \varepsilon \leq 1$.

(i) Show that f may not be absolutely continuous on $[0, 1]$

(ii) Show that f is absolutely continuous on $[0, 1]$ if it is increasing.

Pf. (i) 考虑

$$f(x) = \begin{cases} 0 & x=0 \\ \sin \frac{1}{x} & x \in (0, 1] \end{cases}$$

~2~

在 $[\varepsilon, 1]$ 上

$$|f'(x)| = \left| \frac{1}{x^2} \cos \frac{1}{x} \right| \leq \frac{1}{\varepsilon^2}$$

从而 $f(x)$ Lipschitz 连续, 当然绝对连续. 但在 $[0, 1]$ 上, 对 $\varepsilon_0 = 1$,
 $\forall \delta > 0$, 存在区间

$$\left(\frac{1}{\frac{3\pi}{2} + 2k\pi}, \frac{1}{\frac{\pi}{2} + 2k\pi} \right) \subseteq (a, b), \quad k = \left[\frac{1}{4\pi\delta} \right] + 1$$

使得

$$\left| \frac{1}{\frac{\pi}{2} + 2k\pi} - \frac{1}{\frac{3\pi}{2} + 2k\pi} \right| < \frac{\pi}{(2k\pi)^2} = \frac{1}{4k^2\pi} < \delta$$

但

$$\left| \sin\left(\frac{3\pi}{2} + 2k\pi\right) - \sin\left(\frac{\pi}{2} + 2k\pi\right) \right| = 2 > 1 = \varepsilon_0$$

故 $f(x)$ 在 $[0, 1]$ 上不 Lipschitz 连续.

(ii) 由于 f 在 $[0, 1]$ 上连续必一致连续, $\forall \varepsilon > 0$, 存在 $\delta_1(\varepsilon) \in (0, 1)$

当 $|x_1 - x_2| \leq \delta_1(\varepsilon)$ 时, $|f(x_1) - f(x_2)| < \frac{\varepsilon}{2}$. 由于 f 在 $[\delta_1(\varepsilon), 1]$ 上
 绝对连续, 存在 $\delta_2(\varepsilon) > 0$, 只要 $\sum_{k=1}^n (b_k - a_k) \leq \delta_2(\varepsilon)$ 不变

$$\sum_{k=1}^n (b_k - a_k) < \delta_2(\varepsilon)$$

就有

$$\sum_{k=1}^n |f(b_k) - f(a_k)| < \frac{\varepsilon}{2}.$$

现在对任意 $(a_k, b_k) \subseteq [0, 1]$ 不变, 使得

$$\sum_{k=1}^n (b_k - a_k) < \delta_2(\varepsilon)$$

不妨设 $k=1, 2, \dots, m-1$ 在 $[0, \delta_1(\varepsilon)]$ 内, $k=m$ 包含 $\delta_1(\varepsilon)$, 而 $k=m+1, \dots, n$
 在 $[\delta_1(\varepsilon), 1]$. 若均不在 $\delta_1(\varepsilon)$ 则同理可证. 注意到

$$|f(b_m) - f(a_m)| \leq |f(b_m) - f(\delta_1(\varepsilon))| + |f(\delta_1(\varepsilon)) - f(a_m)| \quad (1)$$

由于 f 单调上升,

$$\sum_{k=1}^{m-1} |f(b_k) - f(a_k)| + |f(b_m) - f(a_m)| \leq |f(\delta_1(\varepsilon)) - f(a_m)| \leq f(\delta_1(\varepsilon)) - f(a_0). \quad (2)$$

~3~

由于 $|\delta_1(\epsilon) - 0| \leq \delta_1(\epsilon)$, 故上式 $< \frac{\epsilon}{2}$. 又因为

$$\sum_{k=m+1}^n (b_k - a_k) + (b_m - \delta_1(\epsilon)) \leq \sum_{k=1}^n (b_k - a_k) < \delta_2(\epsilon)$$

故

$$\sum_{k=m+1}^n |f(b_k) - f(a_k)| + |f(b_m) - f(\delta_1(\epsilon))| < \frac{\epsilon}{2} \quad (3)$$

结合 (1) (2) (3) 得

$$\sum_{k=1}^n |f(b_k) - f(a_k)| < \epsilon.$$

即 f 在 $[0, 1]$ 上绝对连续.

□

4. (Ch 6. 45). Let f be absolutely continuous on $\mathbb{R}$ and g be absolutely continuous and strictly monotone on $[a, b]$. Show that the composition $f \circ g$ is absolutely continuous on $[a, b]$.

Pf. 对于任意 $\epsilon > 0$, 由 f 绝对连续, 存在 $\delta(\epsilon) > 0$, 只要不是开区间 $(a_k, b_k) \subseteq \mathbb{R}$ 使得

$$\sum_{k=1}^n (b_k - a_k) < \delta(\epsilon)$$

就有

$$\sum_{k=1}^n |f(b_k) - f(a_k)| < \epsilon.$$

由于 g 绝对连续且严格单调, 存在 $\eta(\epsilon) > 0$, 只要不是开区间 $(a'_k, b'_k) \subseteq [a, b]$ 使得

$$\sum_{k=1}^{n'} (b'_k - a'_k) < \eta(\epsilon)$$

就有

$$\sum_{k=1}^{n'} |g(b'_k) - g(a'_k)| < \delta(\epsilon).$$

注意到由 g 严格单调, (a_k', b_k') 不空导致 $(g(a_k'), g(b_k'))$ 不空, 故

$$\sum_{k=1}^n |(f \circ g)(b_k') - (f \circ g)(a_k')| < \varepsilon.$$

这就说明 $f \circ g$ 绝对连续.

□.

5. (Ch 6. 56). Let g be strictly increasing and absolutely continuous on $[a, b]$.

(i) Show that for any open ~~set~~ subset O of (a, b) ,

$$m(g(O)) = \int_O g'(x) dx.$$

(ii) Show that for any G_δ subset E of (a, b) ,

$$m(g(E)) = \int_E g'(x) dx.$$

(iii) Show that for any subset E of $[a, b]$ that has measure 0, its image $g(E)$ also has measure 0, so that

$$m(g(E)) = 0 = \int_E g'(x) dx.$$

(iv) Show that for any measurable subset A of $[a, b]$,

$$m(g(A)) = \int_A g'(x) dx.$$

Pf. (i). 由于 $O \subseteq \mathbb{R}$ 可分解为可数个不交开区间之并 (至多集)

$$O = \bigcup_{k=1}^{\infty} (a_k, b_k) \subseteq (a, b).$$

由测度和积分的可加性以及 g 的严格单调性

$$m(g(\bigcup_{k=1}^{\infty} (a_k, b_k))) = m(\bigcup_{k=1}^{\infty} (g(a_k), g(b_k))) = \sum_{k=1}^{\infty} (g(b_k) - g(a_k)).$$

$$\int_O g'(x) dx = \sum_{k=1}^{\infty} \int_{a_k}^{b_k} g'(x) dx.$$

由 Lebesgue 积分基本定理

$$\int_b g'(x) dx = \sum_{k=1}^{\infty} (g(b_k) - g(a_k)) = m(g(\mathcal{O})).$$

(ii). 由定义存在一列开集 $\{\mathcal{O}_k\}_{k=1}^{\infty}$, 使得 $E = \bigcap_{k=1}^{\infty} \mathcal{O}_k$. 令 $\mathcal{U}_n = \bigcap_{k=1}^n \mathcal{O}_k$.

则 $\{\mathcal{U}_n\}_{n=1}^{\infty}$ 单调下降且 $E = \bigcap_{n=1}^{\infty} \mathcal{U}_n$. 由测度和积分的连续性

$$m(g(E)) = m(g(\bigcap_{n=1}^{\infty} \mathcal{U}_n)) = m(\bigcap_{n=1}^{\infty} g(\mathcal{U}_n)) = \lim_{n \rightarrow \infty} m(g(\mathcal{U}_n)).$$

$$\int_E g'(x) dx = \lim_{n \rightarrow \infty} \int_{\mathcal{U}_n} g'(x) dx = \lim_{n \rightarrow \infty} m(g(\mathcal{U}_n)).$$

即知两式相等.

(iii). 对任意 $\varepsilon > 0$, 由绝对连续存在 $\delta(\varepsilon) > 0$, 只要不过开区间 $(a_k, b_k) \subseteq [a, b]$ 满足 $\sum_{k=1}^n (b_k - a_k) < \delta(\varepsilon)$ 就有 $\sum_{k=1}^n |g(b_k) - g(a_k)| < \varepsilon$. 由 E 零测, 存在开集 $\mathcal{O}$ (必要时可加上 (a, b)) 使得 $E \setminus \{a, b\} \subseteq \mathcal{O} \subseteq (a, b)$ 且 $m(\mathcal{O}) < \delta(\varepsilon)$. 由于 $\mathcal{O} \subseteq \mathbb{R}$ 可分解为可数个不过开区间 $(a_k, b_k)_{k=1}^{\infty}$ 之并. 则对任意 $n \in \mathbb{N}^*$,

$$\sum_{k=1}^n (b_k - a_k) \leq \sum_{k=1}^{\infty} (b_k - a_k) = m(\mathcal{O}) < \delta(\varepsilon).$$

于是由 g 严格单调上升, 有

$$\sum_{k=1}^n (g(b_k) - g(a_k)) < \varepsilon$$

~~令 $\varepsilon \rightarrow 0^+$, 由于 $E \setminus \{a, b\} \subseteq \mathcal{O}$ 及测度的单调性.~~

~~令 $n \rightarrow \infty$, 再令 $\varepsilon \rightarrow 0^+$ 得~~

$$\sum_{k=1}^{\infty} (g(b_k) - g(a_k)) = 0$$

由 $E \setminus \{a, b\} \subseteq \mathcal{O}$, 利用测度的单调性

$$m(g(E)) = m(g(E \setminus \{a, b\})) \leq m(g(\mathcal{O})) = \sum_{k=1}^{\infty} (g(b_k) - g(a_k))$$

即知 $m(g(E)) = 0$. 由 Lebesgue 积分的定义, 在零测集 E 上简单函数, 从而可测非负函数, 从而任意可测函数积分都为 0. 于是

$$m(g(E)) = 0 = \int_E g'(x) dx.$$

(iv). 由可测集的性质, 存在 G_δ 集 G , 使得 $A \subseteq G \subseteq [a, b]$ (必要时可支上 $[a, b]$), 且 $m(A) = m(\overline{G})$, 即 $m(G \setminus A) = 0$. 于是由可加性和测度有限,

$$m(g(A)) = m(g(G) \setminus g(G \setminus A)) = m(g(G)) - m(g(G \setminus A))$$

由 (ii) 和 (iii) 的结果

$$m(g(A)) = \int_a g'(x) dx - \int_{G \setminus A} g'(x) dx = \int_A g'(x) dx.$$

□.